

PERSPECTIVE



Hemodynamic precision to guide surgical timing for neonates with congenital diaphragmatic hernia: a narrative review

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Despite a near universal approach focused on physiologic stabilization prior to surgical repair of congenital diaphragmatic hernia (CDH), a condition with significant morbidity and mortality, there remains a lack of consensus, or even guidance, on the appropriate timing for diaphragmatic reconstruction. Hemodynamic-directed care has increasingly been incorporated into the post-natal, peri-operative care of CDH, including assessments of pulmonary hypertension and ventricular dysfunction. Herein, we discuss the integration of targeted neonatal echocardiography and hemodynamic-based assessments to guide precision-directed care and inform selection of the optimal surgical repair window in this complex, heterogeneous patient population.

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INTRODUCTION

Congenital diaphragmatic hernia (CDH) impacts ~1 in 3000 neonates and is associated with significant morbidity and mortality, secondary to a triad of pulmonary hypoplasia, pulmonary hypertension, and cardiac dysfunction [1–3]. The past three decades have seen major advances in the care of neonates with CDH including pressure-limited ventilation, improved pulmonary hypertension strategies such as the transition towards more judicious use of inhaled nitric oxide (iNO), increased focus on cardiac dysfunction, and earlier use of high-quality extra-corporeal life support (ECLS) [4–10]. Despite these advances, there remains no consensus regarding the optimal timing of diaphragm repair and marked variability in repair timing exists between centers [11, 12]. The approach to surgical repair has shifted dramatically from a surgical emergency, necessitating early intervention, to a delayed approach which prioritizes pre-operative clinical stabilization [10, 13–15]. In the most recent guidelines from the American, European, and Canadian CDH consortiums [16–18], this adjudication of clinical stability is heavily weighted by a single parameter that is both rigid and physiologically imprecise; namely systemic oxygenation (fraction of inspired oxygen [FiO₂] or oxygenation index [OI]). In this review, we will highlight the merits of a comprehensive multiparametric approach to decision-making which encompasses the variable and complex nature of the underlying cardiopulmonary physiology.

The potential, veiled limitations of this approach are multifactorial. *First*, some degree of right-to-left shunting, whether at the atrial, ductal, or ventricular septal level, is common in neonates with CDH [19] and will contribute to systemic hypoxemia that is not necessarily reflective of the severity of their underlying disease. *Second*, both FiO₂ and mean airway pressure are key drivers of OI yet are not reliable physiologic characteristics per se as they are often set by the treating clinician in the form of floors

(FiO₂) or lung-protective thresholds (OI). *Third*, the difficulty of distilling multiple physiologic processes and management approaches (pulmonary hypoplasia, pulmonary vascular endowment, cardiac function, ventilator management, pulmonary vasodilator therapy) into a single end-variable is challenging if not impossible. *Fourth*, despite these recommendations, a recent survey of CDH centers found that operative indications or specific targets were not delineated in 41% of guidelines [20]. *Finally*, the dynamic nature of cardiorespiratory physiology and the existence of variable hemodynamic phenotypes has not specifically been considered, or investigated, in the identification of the optimal time for surgical repair [16–18].

TNE and comprehensive neonatal hemodynamic evaluations offer a precision- and physiology-based assessment of critically ill neonates. This approach has been shown to improve outcomes in preterm neonates with a patent ductus arteriosus (PDA) and those with hypoxemic respiratory failure [21–23] and its potential use in the care of neonates with CDH was recently reviewed by Surak and colleagues [24]. In this Perspective, we discuss the use of comprehensive hemodynamic evaluation and echocardiography to help transition away rigid, single operative thresholds and toward a focus on *hemodynamic stability* on serial assessments.

EMERGING EVIDENCE

The past decade has witnessed a gradual trend away from utilizing a singular metric of clinical stability. Indeed, the most recent definition of operative readiness by the Canadian CDH collaborative includes an echocardiography threshold of estimated pulmonary pressures [16]. Deeney and colleagues implemented a threshold of echocardiogram derived estimates of pulmonary pressures <80% systemic prior to surgical repair and found that this was associated with reduced post-operative decompensation

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Table 1. Recommended echocardiographic markers of pulmonary pressures and ventricular function in neonates with congenital diaphragmatic hernia (CDH) [28].

Measure	Approach	Interpretation	Benefits	Limitations
Pulmonary Pressures	PDA shunt	- All LR: sub-systemic PH - Bidirectional: $\leq$ systemic PH - All or predominant RL: supra-systemic PH	Accurate estimate of PAP relative to systemic	- Requires presence of PDA - Shunt direction is influenced by both pulmonary and systemic pressures
	TR jet	- RVSp (plus RA pressure) > 30 mmHg indicates PH of any etiology	- Ordinal - Can be indexed directly to systolic BP	- Requires presence of complete TR jet - Will be under-estimated with RV dysfunction
	Septal position	Qualitative - Flat: $\geq \frac{1}{2}$ systemic PH - Paradoxical: supra-systemic PH Quantitative (sEI) - ≥ 1.3 : $\geq \frac{1}{2}$ systemic PH	- Always present	- Influenced by systemic blood pressure (under-estimated if high systemic BP) - Very poor interrater agreement (qualitative) [37]
RV Function	TAPSE	- Estimate of RV contractility - GA-specific thresholds [89]	- Reproducible measure of RV systolic function	- Highly angle-dependent - Only systolic function
	FAC	- ≥ 0.35 suggests normal RV function	- Assesses global RV function	- Can be difficult to measure with CDH anatomy
LV Function	EF	- EF $\geq 55\%$ by Simpson's Biplane suggests normal LV function	- Measure of LV systolic function	- RV septal bowing can distort anatomy
RV and LV Function	TDI	- Measure of ventricular systolic and diastolic velocities - GA-specific thresholds [90]	- Assesses global RV and LV function	- Angle-dependent
	Strain / Speckle Tracking	- Normative data in CDH requires further research	- May be more sensitive than EF [43]	- Requires experienced operator

PDA patent ductus arteriosus, LR left-to-right, PH pulmonary hypertension, sEI systolic eccentricity index, RL right-to-left, PAP pulmonary artery pressure, RVSp right ventricular systolic pressure, RA right atrium, BP blood pressure, RV right ventricular, TAPSE tricuspid annular plane systolic excursion, GA gestational age, FAC fractional area change, EF ejection fraction, LV left ventricular, TDI tissue Doppler imaging.

[25]. Similarly, Shinno and colleagues implemented a protocol to pursue operative repair once the patent ductus arteriosus (PDA) shunt was >50% left-to-right and demonstrated reduced duration of mechanical ventilation and a trend towards earlier repair [26]. Whilst these approaches provide additional insights, they are limited by a lack of prospective study and simplifying decision-making to a single measure, pulmonary artery pressure (PAP) alone, rather than underlying and more complex hemodynamic mechanisms.

ASSESSING PULMONARY PRESSURES IN NEONATES WITH CDH CDH-associated pulmonary hypertension (PH) is a key pathophysiologic driver of outcomes in CDH [27]. Echocardiography assessment of pulmonary pressures must be systematic, tiered, and multiparametric [28] (Table 1).

An evidence-based approach [29–32] begins **first tier** with an assessment of PDA shunt direction. While a powerful tool, it is limited by requiring the presence of a PDA. This is near universal in neonates with CDH within the first 48 hours but decreases in incidence with time [33]. Further, PDA shunt direction can be limited by its association with systemic blood pressures wherein a left-to-right shunt will increase with increasing systemic blood pressure and vice versa. Finally, it is important to attempt to quantify the degree of flow directionality as the clinical relevance of a PDA that is 90% left-to-right versus the inverse is markedly different. This approach was recently shown to trend improvement of CDH pulmonary hypertension in two separate studies [26, 33].

The **second tier** involves the calculation of estimated right ventricular systolic pressure (RVSp) that is obtained by measuring the velocity of the tricuspid regurgitant (TR) jet and utilizing the modified Bernoulli equation [34]. While this provides an ordinal value, it can underestimate PH in the setting of an incomplete

regurgitant jet and/or right ventricular (RV) dysfunction [35]. A similar approach can be used if a ventricular septal defect (VSD) is present [25].

The **third tier** involves an assessment of pulmonary pressure by interrogating interventricular septal position. While this determination is often made subjectively [36], Smith and colleagues recently demonstrated poor interrater agreement (Cohen's Kappa = 0.20) for the subjective interpretation of septal position and a high misdiagnosis rate of nearly 1 in 5 [37].

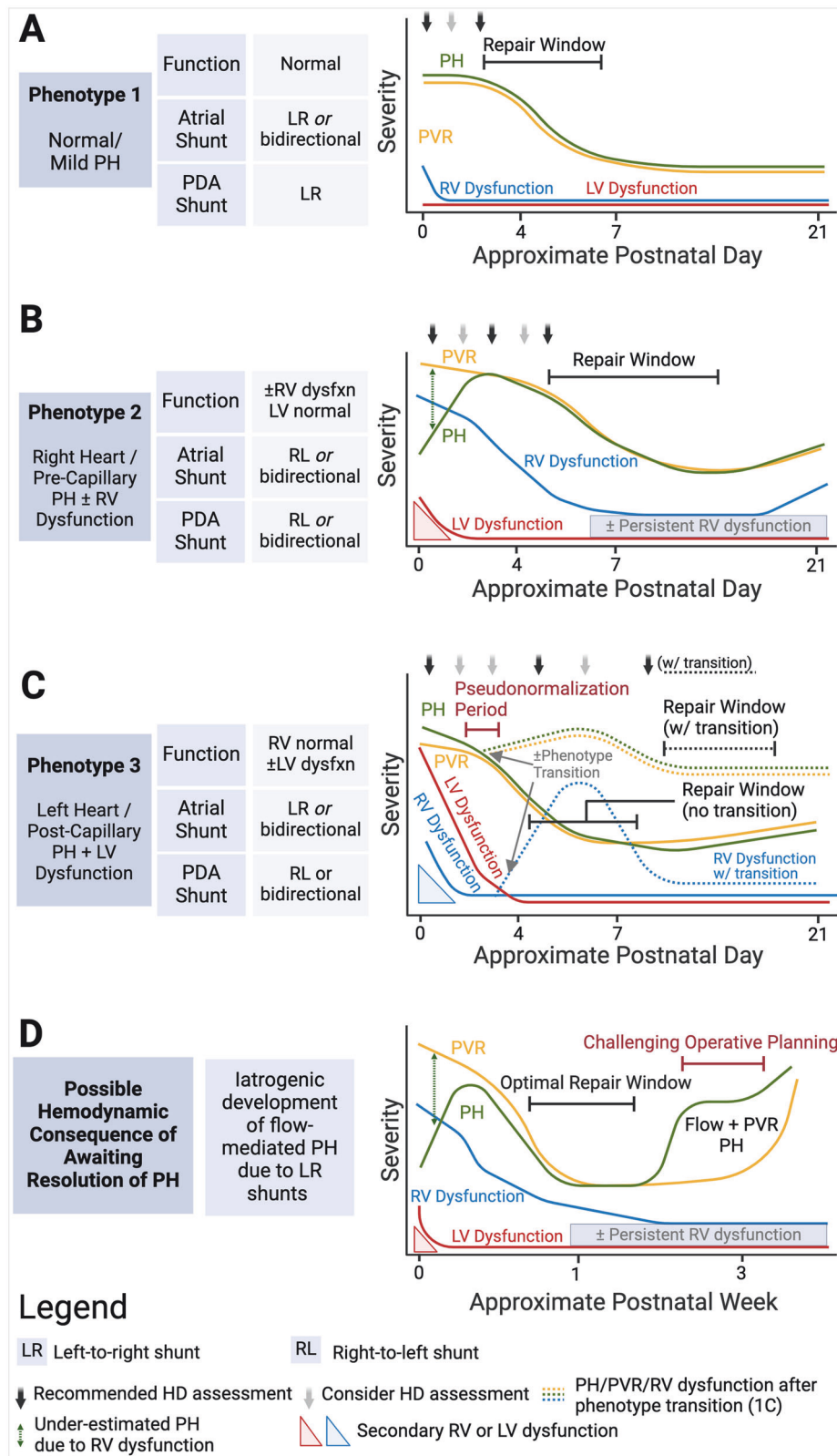
UNIQUE PHYSIOLOGIC CHALLENGE OF INTERPRETING ELEVATIONS IN PULMONARY PRESSURES IN CDH

Interpreting pulmonary pressures in CDH requires a detailed appreciation of their complex physiology. Modifying Ohm's law for physiologic blood flow:

$$\text{Mean PAP} = \text{Pulmonary Vascular Resistance (PVR)} \times \text{Blood Flow} + \text{Left Atrial (LA) pressure}$$

It is important to note that echocardiography estimates of PH reflect only pressure (PAP) and *not* resistance (PVR), which can only reliably be measured via cardiac catheterization. This is illustrated by the divergence of the PH and PVR lines in various components of Fig. 1.

Incorporating Ohm's law to CDH physiology: *first*, the expected decrease in PVR after birth is interrupted. Typically driven by oxygen exposure, lung inflation, and rising systemic blood pressure [38, 39]; the relative hypoxemia, pulmonary hypoplasia and intra-thoracic abdominal visceral competition, and cardiac dysfunction inherent in CDH will result in a delayed PVR nadir, reflected in Fig. 1B [40, 41]. *Second*, intra-/extra-cardiac shunts can alter pulmonary blood flow; specifically, any component of left-to-



right shunting will increase pulmonary blood flow and, by Ohm's law, potentially increase PAP (Fig. 1D). With the growing use of Prostaglandin E1 to maintain ductal patency in severe CDH [42], this is key to emphasize as decreases in PVR may lead to an improvement in left-to-right shunting which can then paradoxically contribute to a persistent elevation in PAP. Finally, left

ventricular systolic and diastolic dysfunction is common in CDH [8] and can lead to LA hypertension and up-stream post-capillary elevation in PAP (Fig. 1C).

Taken together, as with FiO_2 , predicted PAP provides only a single measure of clinical stability and should not be relied on in isolation.

Fig. 1 Illustrative examples of common hemodynamic phenotypes in CDH and their relationship to optimal surgical repair timing. These phenotypes are not exhaustive, and not all neonates will fit into one distinct category, yet they represent repeatable patterns. As discussed in the text, “Repair Window” is defined as a composite of (1) clinical stability based on institution-specific targets (oxygenation, ventilation, etc.), (2) normal or resolution of left ventricular (LV) dysfunction, (3) optimization of pulmonary vascular resistance (PVR) and normalization or stability of right ventricular (RV) function. **A** No/mild pulmonary hypertension (PH) and essentially normal biventricular function. Very early, mild RV systolic dysfunction is likely to be transient and can be followed serially and should not delay operative repair. **B** Pre-capillary PH +/- right ventricular (RV) dysfunction. RV dysfunction will likely be slower to completely normalize, and attention should be focused on stability/absence of worsening on serial hemodynamic (HD) assessments. **C** Post-capillary PH with left ventricular (LV) dysfunction. Critically, LV dysfunction will appear and often correct sooner but can mask concomitant RV dysfunction. It is important to ensure biventricular function has stabilized (avoiding a pseudo-normalization period where RV dysfunction is persistent/increasing as the LV improves [denoted by the hatched lines]) prior to surgical repair. Of note, even without this transition to RV-predominant dysfunction, the surgical repair window is delayed relative to Phenotype 1 due to the severity of initial LV dysfunction and need for observation of phenotype transition. **D** A hemodynamic consequence of prolonged surgical delay where heart function normalizes and PVR falls such that the nature of PH (flow-mediated) transitions (and worsens) to a phenotype secondary to prolonged left-to-right (LR) shunts PH. Additionally, PDA, patent ductus arteriosus; RL, right-to-left shunt; PAP, pulmonary artery pressures. Figure created in <https://Biorender.com>.

ASSESSING CARDIAC FUNCTION IN NEONATES WITH CDH

Similar to PAP, a CDH hemodynamic assessment should include a quantitative multi-parametric assessment of biventricular function. As seen in Table 1, RV function should be assessed by, at minimum, tricuspid annular plane systolic excursion (TAPSE), fractional area change, and tissue Doppler imaging (TDI) [24, 28, 37]. Strain analysis via speckle tracking may be even more sensitive but requires significant center expertise and experience [43]. Left ventricular (LV) function can be assessed by ejection fraction via Simpson's Biplane, TDI, and speckle tracking/strain [44]. In evaluating RV or LV systolic function, a multi-parametric assessment is recommended to mitigate the impact of inter-/intra-observer variability of individual parameters. Consistency of results increases the confidence in adjudicating heart dysfunction. Finally, strong consideration should be given towards utilizing serial measures of left and right ventricular outputs particularly in regard to shunt volume interpretations [28].

CURRENT USE OF HEMODYNAMIC ASSESSMENTS IN CDH

Hemodynamic-directed care has already been used to improve the management of CDH-associated PH by identifying patients who may benefit from iNO leading to a potential reduction in need for ECLS [4]. Similarly, this approach has been utilized for CDH-associated cardiac dysfunction. Patel and colleagues identified cardiac dysfunction in as many as 58% of neonates in the first 48 hours [35] and which was independently associated with need for ECLS [8] and mortality [7]. Finally, a recent study by Kuan et al., there was a reduction in mortality following implementation of a hemodynamic-focused CDH care bundle [45].

CARDIOVASCULAR PHENOTYPES IN CDH

With the growing utilization of echocardiography assessments of CDH over the past decade, there is a growing appreciation of distinct hemodynamic phenotypes in CDH involving both cardiac function and PAP/PH [1, 46]. As seen in Fig. 1, these are generally considered to be delineated echocardiographically into three broad categories. The *first category* (1) includes those with no/mild PH and normal cardiac function. While both PVR and PAP are likely elevated [33], they are clinically well-tolerated with preserved ventricular function and PAP remains systemic to sub-systemic. The *second category* (2) is those with pre-capillary PH with or without RV dysfunction. In this sub-group, PVR is markedly elevated leading directly to a significant elevation in PAP. Depending upon pressure-loading conditions among others, there may be either RV or biventricular with RV predominant dysfunction and/or RV dilation. The *third category* (3) is those with post-capillary PH and LV dysfunction [46]. In this sub-group, LV dysfunction (systolic and/or diastolic) [8] leads to up-stream LA hypertension and back-pressure eventually transmitted through

the pulmonary vessels back to the right heart. While PVR may be elevated in this sub-group secondary to *in utero* pulmonary vascular maldevelopment, the PAP elevation is disproportionately driven by elevated LA pressure from LV dysfunction.

TOWARDS A MULTI-PARAMETRIC HEMODYNAMICS-DIRECTED ASSESSMENT OF SURGICAL TIMING

The range of utilization of echocardiography assessments varies by both center and timing. Regarding consensus approaches, the Extracorporeal Life Support Organization (ELSO) guidelines suggest an initial echocardiogram assessment at 4–12 hours of life [47] while the CDH Study Group (CDHSG) records an early postnatal, a preoperative, and a near discharge echocardiogram (reflecting common baseline practices) [48]. At centers with established hemodynamic programs, near daily (or even more frequent) comprehensive TNE and Neonatal Hemodynamics consultations [49, 50] are the norm in the pre-repair window [24, 45]. These early and serial TNE assessments enable full identification and appreciation of initial hemodynamic phenotypes and their dynamic course in individual patients to guide all aspects of clinical decision-making. At other centers where this service is not available, comprehensive hemodynamic assessment, including echocardiogram, should be performed *at minimum* once after birth and again after a threshold of clinical stability has been achieved before surgical correction of the diaphragmatic hernia (Fig. 2).

This first-line assessment of surgical readiness must be a daily assessment of clinical status by the Neonatology and Pediatric Surgical team and should follow the consensus and institution-specific guidelines of markers including systemic oxygenation, blood gas values, and tissue/end-organ perfusion [16–18].

Rather than focusing exclusively on single time-point, specific mandated, and physiologically imprecise thresholds (OI, FiO₂, etc.), we endorse keying on clinical stability over time to trigger a hemodynamic assessment of operative readiness.

HEMODYNAMIC ASSESSMENT OF OPERATIVE READINESS

Right and left ventricular function

Cardiac dysfunction in CDH is common and has a significant impact on morbidity, ECLS use, and mortality [7, 8, 51, 52]. Patel and colleagues identified RV dysfunction in 34% and LV dysfunction in 24% of neonates with CDH in the first 48 hours [7], while others have observed RV and LV dysfunction in 56–66% and 40–52% of cases, respectively [53]. Incorporating advanced echocardiographic assessments (strain), ventricular dysfunction may be even more common, as Fraga and colleagues recently identified LV dysfunction, that was associated with ECLS, in >60% of neonates with CDH [54].

Less is known regarding the *timeline or natural history* of cardiac dysfunction. Very mild RV systolic dysfunction has been noted by

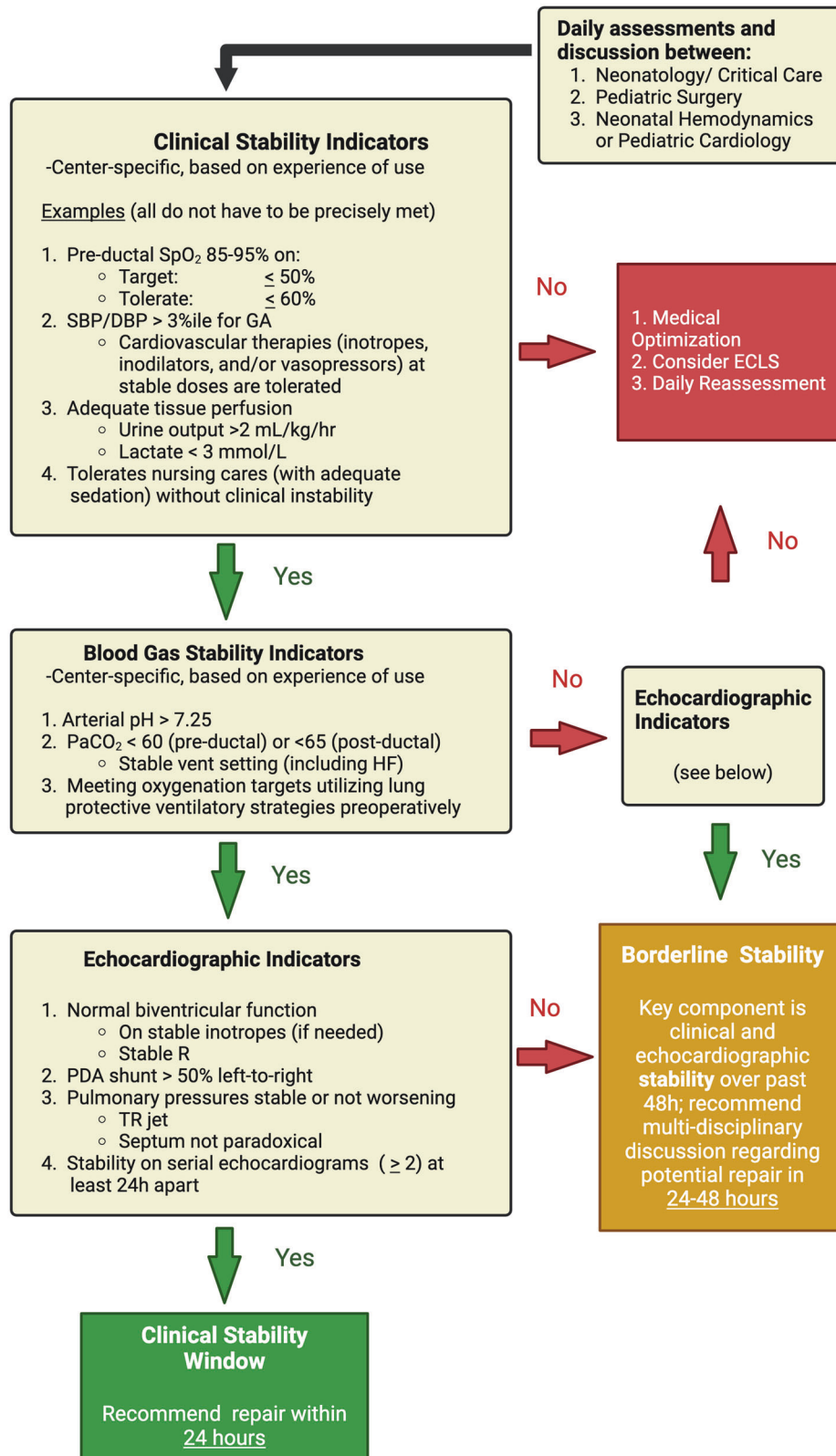


Fig. 2 Hemodynamic-guided algorithm to surgical repair timing in neonates with CDH. Proposed algorithm incorporating clinical stability, arterial blood gas, and echocardiographic indicators. SpO₂ peripheral oxygen saturation, BP blood pressure, GA gestational age, PaCO₂, partial pressure of carbon dioxide in arterial blood, HF high frequency ventilation, including high frequency oscillatory ventilation and high frequency jet ventilation, PDA patent ductus arteriosus, RV right ventricular, TR tricuspid regurgitation, ECLS extracorporeal membrane oxygenation. Figure created in <https://Biorender.com>.

the authors on screening TNE soon after delivery that is transient and needs only serial observation (*unpublished observations*, Fig. 1A). This likely reflects an exaggerated response in CDH of the normal transient reduction in RV function observed following cord-clamping [55]. Persistent RV dysfunction appears to be predominantly diastolic in the early course of CDH and improves in the first 72–96 hours in those not requiring ECLS [56]. RV systolic dysfunction appears to follow a similar course (Fig. 1B) normalizing at a median of 2 days in one study [33]; however, RV dysfunction persisted post-repair in a subset of patients up to 30 days while persistent RV diastolic dysfunction has been seen up to 5 years post-repair [43, 57]. Serial echocardiographic assessments are helpful to distinguish mild transient RV dysfunction (Fig. 1A), which is not a contra-indication to surgery, from more severe RV dysfunction which may delay repair (Fig. 1B). LV function appears to correct faster (Fig. 1C), mostly within the first 72 hours [33, 35, 58]. It is important to note that heart dysfunction is often not univentricular and, due to ventriculo-ventricular interactions and/or ventricular interdependence [59], LV dysfunction can be intermixed with RV dysfunction (as seen in Fig. 1B and C). Indeed, Patel and colleagues observed biventricular dysfunction in 42% of cases [35]. As LV dysfunction tends to correct first, it is possible that a predominant LV phenotype (Fig. 1C) transitions to an ongoing RV dysfunction phenotype. A period of relative *pseudo-normalization* of function may therefore occur where improving LV function and a reduction in LA pressure leads to a brief drop in PAP while PVR continues to rise insidiously leading eventually to worsening of PAP and RV dysfunction (Fig. 1C, **dotted lines**). Clinically, this may be observed by improving markers of systemic perfusion whilst systemic oxygenation worsens. During this window of transition, we caution against intervening surgically if RV function is not stabilizing or improving.

Taken together, surgical repair in neonates with CDH not receiving ECLS must be timed carefully with the expected normalization of cardiac dysfunction either via time or medications. In most cases, this will be after **72–96 hours**. Underlying dysfunction may persist particularly for the RV (Fig. 1B) and thus the key is *stability* on serial assessments. Finally, given the tight association of heart function with outcomes in CDH, cardiotropic medications, should not be discontinued prior to operative repair.

Pulmonary Vascular Resistance

Due to both embryogenic alterations and pulmonary vascular remodeling *in utero*, PVR will be both higher and slower to improve in CDH as compared to neonates without CDH [60, 61]. As discussed above, PVR is not synonymous with PAP/PH. It is largely linearly related in Phenotypes 1 and 2 (Figs. 1A and 1B) where changes in PVR do generally directly drive PH with one key exception. As seen in Fig. 1B (**hatched arrow**), RV systolic dysfunction will lead to an apparent lower PAP that may paradoxically increase as RV function improves, either with time or inotropes. This illustrates the importance of serial assessments to fully capture the changes in PVR in CDH. In Phenotype 3 (Fig. 1C), the reduction in PVR will *not* be correlated with PAP and will likely in fact be inversely related with increasing PH secondary to excess pulmonary blood flow.

Optimally, surgical correction would occur with the **nadir of PVR**; however, this can only be precisely measured by cardiac catheterization which is both invasive and impractical. As such, we do not recommend attempting to use a quantifiable estimate of PVR but rather a comprehensive assessment of the patient's physiology and characterization of the dominant phenotype. In select cases, pulmonary arterial pressures can be used as a surrogate for PVR *if* (and only *if*) there is a pure pre-capillary PH phenotype *and* after correction of any RV dysfunction (Fig. 1B). In the authors' experience, however, there is frequent intermixing of these phenotypes.

Pulmonary hypertension

Unlike ventricular function, we **should not expect or require** normalization of PAP/PH prior to surgical repair. Indeed, as depicted in Fig. 1, the optimal repair window of all cases includes some degree of stable or not worsening PH. Indeed, multiple catheter and echocardiogram-based studies have documented persistence of PH at the time of repair and beyond [62, 63]. It is important, however, to target the nadir of existing PH prior to surgical repair as PAP < 1/2 systemic has been associated with improved outcomes and supra-systemic PAP is associated with mortality [64].

Taken together, there are competing targets of (1) anticipating that PH will not necessarily resolve in all cases pre-repair and (2) optimizing PH management recognizing that there will be hemodynamic fluidity in the pre-repair window. To address this challenge, we recommend a paradigm shift away from a sole focus on the presence or severity of PH and towards a focus on (1) *well-controlled PH* and (2) *hemodynamic stability* on serial assessments as depicted in Fig. 2. This approach enables both identification of a repair window where clinical and pharmacologic management has optimized PAP and elucidates the true underlying hemodynamic phenotype.

IMPORTANCE OF EARLY IDENTIFICATION OF REPAIR WINDOW

While it is critical to delay surgical repair until clinical stabilization, *it is equally as important to not postpone repair when acceptable stabilization has been achieved*. Data from the CDHSG, revealed that per-day changes in surgical repair date were associated with an increase in post-repair ventilator days, need for oxygen and NG feeds at discharge [65]. Further, a single-center study from a high-volume CDH center found that per-day delays in surgical repair, once a stability threshold was met, was associated with an increase in ventilatory days and total hospital duration [66]. As depicted in Fig. 1D, postponing surgical repair after optimal pre-capillary PH has been managed can lead to worsening PH secondary to excessive pulmonary blood flow from high volume left-to-right shunts (specifically those patients with a persistent or PGE-maintained PDA). In essence, we must “not let perfection be the enemy of good.”

SPECIAL CONSIDERATIONS

While the primary focus of this review relates to considerations from a hemodynamic/echocardiography perspective, it is important for the readers to recognize the importance of a comprehensive patient evaluation when seeking optimal timing for diaphragmatic repair. Additional factors which frequently influence cogitation regarding surgical readiness in CDH include *gestational age (GA)*, *weight*, *anatomic cardiac anomalies (congenital heart disease [CHD])*, *multiorgan failure*, *presence of severe encephalopathy and chromosomal anomalies*. While a thorough discussion of each factor is outside of the scope of this review, these factors complicate the discussion of the optimal window for repair. The approach to definitive repair in preterm infants is particularly challenging. Of note, in preterm infants, once a GA threshold of >32 weeks and weight >1.8 kg are reached, ECLS is now feasible [47, 67, 68]. On the contrary, waiting for premature or low birth weight patients to reach these thresholds may miss an opportunity to (1) optimize lung maturation in the late canalicular or early sacellar phases of lung development and (2) initiate early enteral nutrition. CDH + CHD and CDH + chromosomal/genetic anomalies further complicate an already highly complex situation, including both pathophysiologic nuance and ethical / family considerations, and are addressed elsewhere [69, 70].

HEMODYNAMIC CHANGES DURING CDH REPAIR

Achieving optimal clinical stability prior to surgical repair is necessary due to the significant hemodynamic challenges

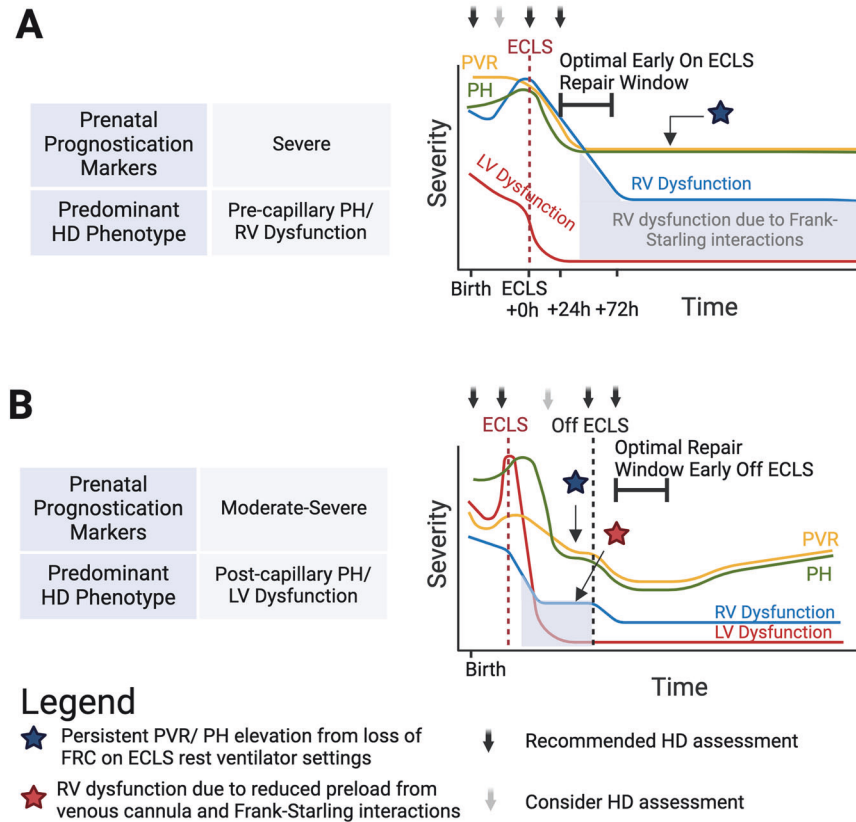


Fig. 3 Case examples of CDH phenotypes requiring ECLS and their relationship to hemodynamic-guided optimal surgical repair timing. Of note, these represent example cases and not all pre-capillary phenotypes will fall exactly into a specific case. In both examples the window between birth and ECLS is usually 1–48 hours. **A** Predominant pre-capillary pulmonary hypertension (PH) and right ventricular (RV) dysfunction phenotype with severe pre- and peri-natal risk stratification that is unlikely to rapidly resolve indicating recommended early on-extracorporeal life support (ECLS) repair. **B** Predominant post-capillary PH with left ventricular (LV) dysfunction with more moderate-severe pre- and peri-natal risk stratification that is more likely to improve quickly indicating recommended repair after decannulation. Depicted are expected hemodynamic features while on ECLS including some degree of expected persistent RV dysfunction and persistent elevation of pulmonary vascular resistance (PVR) and PH (described in text). IUGR intra-uterine growth restriction, *Honeymoon period*, denotes a period of critical illness stability/ improvement seen soon after birth [91, 92]; HD hemodynamic, AGA appropriate for gestational age. Figure created in <https://Biorender.com>.

encountered during the procedure. Manipulation of the liver, bowel, and diaphragm has been shown to increase PVR and PAP [63]. Further, both neuromuscular blockade and carbon dioxide (CO₂) insufflation during thoroscopic repair are associated with hypercarbia which increases pulmonary vasoreactivity [71]. A study by Dotta and colleagues using near infrared spectroscopy found that, despite adequate peripheral oxygen saturations, cerebral oxygenation was reduced during abdominal visceral reduction likely secondary to impaired venous return [72]. While further work is needed, these highlight the need for tight and pre-emptive peri-operative management.

INCORPORATING HEMODYNAMIC PRECISION INTO THE TIMING OF CDH REPAIR ON ECLS

Despite the elevated mortality when utilizing ECLS [73], there is a clear indication and improved outcome specifically for neonates with severe CDH. In this subset, ECLS has been shown to improve survival relative to those managed medically [74, 75]. Rapid identification of this severe CDH phenotype can be achieved via a hemodynamics-based assessment, specifically with regards to persistent ventricular dysfunction [33] and hemodynamic estimates of pulmonary vascular endowment and pressures [76–79].

There has been no established consensus to guide surgical timing for infants with CDH on ECLS. Historically there was a trend towards opting for repair after decannulation to minimize

bleeding complications and morbidity associated with on-ECLS repair [80, 81]. However, growing evidence in the past decade appears to suggest a likely benefit to early on-ECLS repair [82]. The evidence of benefit is most pronounced for those neonates with the most severe phenotype and/or the highest overall risk (as assessed by prenatal imaging, hemodynamic severity, or other strategies) [82–84]. Conversely, those neonates with an anticipated short duration ECLS run may benefit from delaying surgical repair until after decannulation [81] (Fig. 3). A hemodynamics-based assessment can aid in identifying clinical stability on ECLS for these early repairs and in predicting those with the potential for liberation from ECLS for an off-ECLS repair. While a full discussion of this is beyond the scope of this perspective, there are key hemodynamic parameters to highlight.

(1) Normalization of LV function is an important consideration prior to repair on ECLS or decannulation for repair. While we recommend awaiting improvement of LV dysfunction with either improved oxygenation following veno-arterial (VA) ECLS or initiation of inotropes prior to repair, normalization of LV function is absolutely required for those on veno-venous (VV) ECLS. Conversely, mild-moderate RV dysfunction may persist on both VV and VA ECLS secondary to emptying of the RV from the venous cannula and expected impaired contractility due to Frank-Starling interactions [85] and should not by itself delay surgical repair (Fig. 3A).

(2) While there is significant center-to-center variability in the use of pulmonary vasodilators while on VA ECLS, it is important to

recognize that evidence of PH should not preclude on-ECLS repair. High pressure ventilation as a strategy to optimize oxygenation prior to ECLS should be avoided as it may further compromise low pulmonary blood flow either directly through its effects on increasing PVR or indirectly through increased RV afterload. Indeed, in cases where excessive peri-cannulation mean airway pressure is employed, reducing mean airway pressure while on ECLS will improve PVR by transitioning the neonate toward functional residual capacity (FRC). Conversely, in many centers respiratory support will be weaned on ECLS away from FRC inducing de-recruitment [86]. This will in turn, based on the lung hysteresis curve, increase PVR and thereby increase PH [87]. Thus, we do not recommend awaiting normalization of pulmonary pressures on ECLS prior to surgical repair. Rather, we advocate again for *stability on serial hemodynamic assessments* (Fig. 3A).

(3) A pre-ECLS hemodynamic assessment suggesting a post-capillary PH phenotype with LV dysfunction coupled with moderate-severe perinatal risk factors and/or an initial period of clinical stability ('honeymoon period') in the authors' experience is likely to correct quickly after ECLS support (within 12 hours). This can be confirmed with a TNE assessment on ECLS showing normalizing LV function (Fig. 3B) and is suggestive of two possibilities: 1) appropriate candidacy for early repair or 2) appropriate for early weaning of ECLS support, decannulation, and repair after decannulation. This approach requires accepting the risk of a late repair on ECLS if liberation is unsuccessful. Further work is needed to investigate hemodynamic assessments on full ECLS support.

INCORPORATING HEMODYNAMIC ASSESSMENTS IN NON-TNE CENTERS

The use of TNE is growing across North America and Europe [88], however it is still present in a minority of centers. A comprehensive hemodynamic assessment of neonates with CDH would ideally but not exclusively incorporate TNE. More important by far than who performs the echocardiogram is (1) a focus on acquiring the key, quantitative measures (Table 1), (2) interpreting these in the context of cardiopulmonary physiology with particular attention paid to Ohm's Law, and (3) the need for serial, high-quality assessments which are integrated within the specific clinical context. There is a need for growing multi-center research and discussion regarding incorporating CDH hemodynamic phenotypes into the direct care and surgical timing decisions of these patients.

CONCLUSIONS

The approach to surgical repair timing in CDH is complex and multi-factorial. A hemodynamics assessment that uses comprehensive, multiparametric and quantitative echocardiography has the potential to provide precision-based care to optimize surgical timing by identifying a period of *stability on serial cardiopulmonary examination* rather than a priori clinical targets. We endorse consideration for the use of hemodynamic assessment to guide timing of surgery, based on improvement in ventricular function and optimization, rather than resolution, of pulmonary hypertension. Further investigation is needed to further characterize and evaluate this approach, and multi-center prospective collaboration should add evidence in the coming years.

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AUTHOR CONTRIBUTIONS

JTW and PJM developed the idea of the manuscript, JTW, NP, MTH, and PJM outlined the nature, extent, and direction of the perspective, JTW wrote the manuscript and created the figures, JTW, NP, MTH, and PJM edited and revised the final document.

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